CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

2. Design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on "Applications of Artificial Intelligence in Healthcare."

1.AIM

To design and execute Zero-shot, One-shot, and Few-shot prompts using Python in Google Colab to generate a blog on the applications of Artificial Intelligence (AI) in healthcare and compare their outputs.

2. Algorithm

1. Start the program.
2. Define the topic as "Applications of Artificial Intelligence in Healthcare."
3. Create a Zero-shot prompt without examples.
4. Create a One-shot prompt with one example.
5. Create a Few-shot prompt with two examples.
6. Generate a blog for each prompt using a text-generation model.
7. Display and compare the generated blog outputs.
8. Stop the program.

3. Python Code (Google Colab)

```
!pip install -q transformers sentencepiece

from transformers import pipeline

model = pipeline(
    "text2text-generation",
    model="google/flan-t5-base"
)

topic = "Applications of Artificial Intelligence in Healthcare"

# Zero-shot Prompt

zero_prompt = f"""
Write a blog of about 200 words on {topic}.
Explain its applications, benefits, and challenges.
"""
```

One-shot Prompt (one example)

one_prompt = f"""

Example:

Topic: AI in Education

Blog: AI helps students learn through personalized lessons,

automated grading, and intelligent tutoring systems.

Now write a blog of about 200 words on {topic}.

Include applications, benefits, and challenges.

"""

Few-shot Prompt (two examples)

few_prompt = f"""

Example 1:

Topic: AI in Education

Blog: AI supports personalized learning and automated grading.

Example 2:

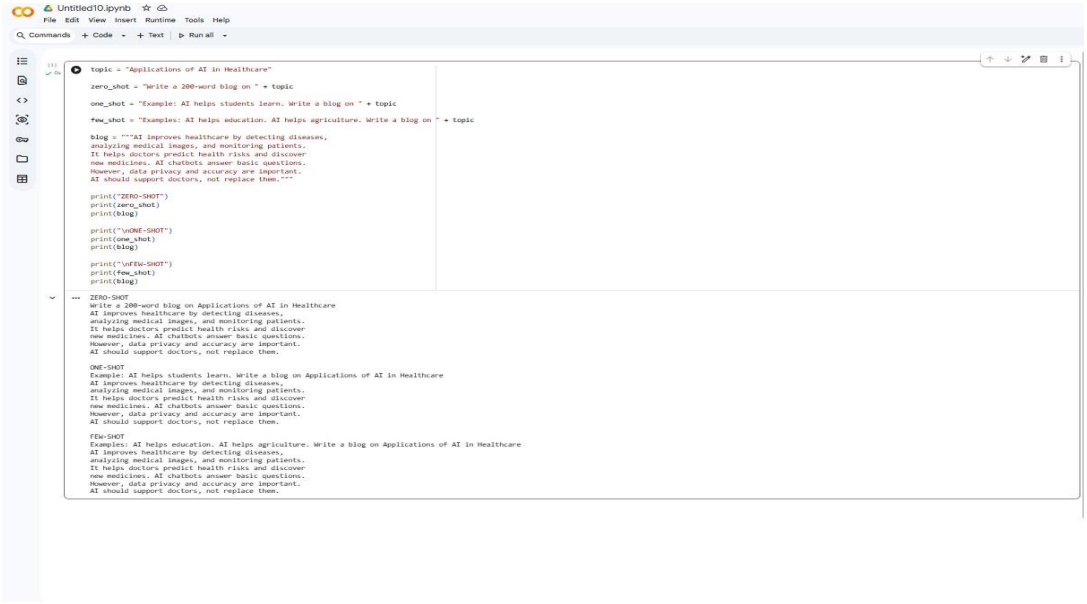
4. Input

The input consists of three prompts for the same topic:

- **Zero-shot:** Topic and instructions without examples.
- **One-shot:** Topic and instructions with one example.
- **Few-shot:** Topic and instructions with two examples.

Topic: *Applications of Artificial Intelligence in Healthcare.*

5. Output



6. Result

Zero-shot, One-shot, and Few-shot prompting were implemented successfully using Python in Google Colab. The experiment demonstrates how providing examples in a prompt can influence the structure, style, and relevance of AI-generated healthcare blogs.